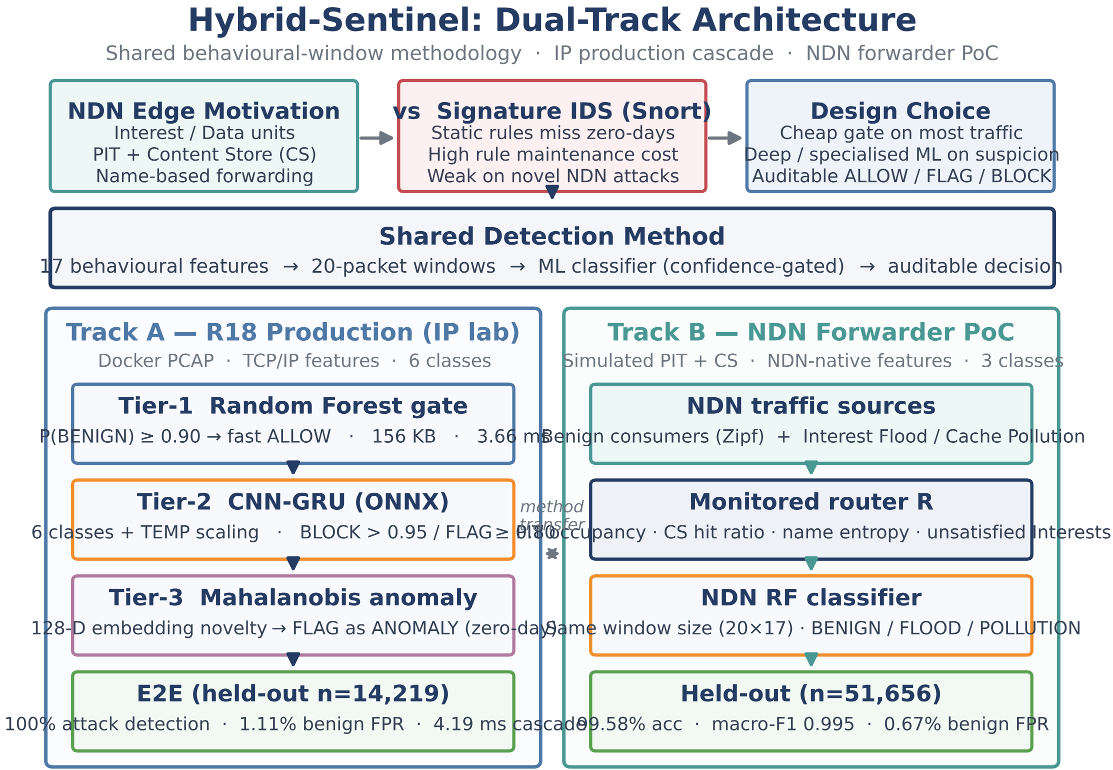
Hybrid-Sentinel: Dual-Track Architecture
Shared behavioural-window methodology · IP production cascade · NDN forwarder PoC
NDN Edge Motivation
Interest / Data units
PIT + Content Store (CS)
Name-based forwarding
vs Signature IDS (Snort)
Static rules miss zero-days
High rule maintenance cost
Weak on novel NDN attacks
Design Choice
Cheap gate on most traffic
Deep / specialised ML on suspicion
Auditable ALLOW / FLAG / BLOCK
Shared Detection Method
17 behavioural features → 20-packet windows → ML classifier (confidence-gated) → auditable decision
Track A — R18 Production (IP lab)
Docker PCAP · TCP/IP features · 6 classes
Tier-1 Random Forest gate
P(BENIGN) ≥ 0.90 → fast ALLOW · 156 KB · 3.66 ms
Tier-2 CNN-GRU (ONNX)
6 classes + TEMP scaling · BLOCK > 0.95 / FLAG ≥ 0.80
Tier-3 Mahalanobis anomaly
128-D embedding novelty → FLAG as ANOMALY (zero-day)
E2E (held-out n=14,219)
100% attack detection · 1.11% benign FPR · 4.19 ms cascade
method transfer
Track B — NDN Forwarder PoC
Simulated PIT + CS · NDN-native features · 3 classes
NDN traffic sources
Benign consumers (Zipf) + Interest Flood / Cache Pollution
Monitored router R
PIT occupancy · CS hit ratio · name entropy · unsatisfied Interests
NDN RF classifier
Same window size (20×17) · BENIGN / FLOOD / POLLUTION
Held-out (n=51,656)
99.58% acc · macro-F1 0.995 · 0.67% benign FPR